

Image Resolution Effects on Deep Learning Models for Video-Assisted Intubation

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Abstract

In this study, we investigate the impact of image resolution on the performance of Deep Learning (DL) models for classifying video-assisted endotracheal intubation (ETI) assessments as a pass or fail. This study utilized 50 balanced laryngoscopy videos, annotating key anatomical structures with polygon shapes using LabelMe. A 3D CNN and a hybrid 2D CNN for feature extraction with an SVM classifier were trained and tested using 4-fold cross-validation across 5 different input resolutions (64x64, 128x128, 224x224, 320x320, and 600x600). In order to assess the performance of this model, we used metrics such as the mean accuracy across folds, the AUC, the F1-score, and the training duration. The overall results showed that the 2D CNN + SVM model was more consistent across resolution, outperforming the 3D-CNN model in all aspects, achieving an average accuracy of 83.55% between all 5 resolutions while also being 1,825x faster at the higher resolutions. We observed that the 2D CNN + SVM model performed the best at 64x64 and 224x224 resolutions, where training time, accuracy, and other metrics were satisfactory. Throughout this experiment we were able to highlight the importance of resolution selection when training DL models in a medical application, as well as support the use of hybrid architectures for more efficient and deployable ETI classification.

Introduction

Image resolution plays a huge part in the training and testing of any deep learning model(s); however, in most medical applications and deep learning, resolution is sometimes sacrificed to adhere to a fast computational model, therefore, eliminating details in these images that would otherwise be important [1]. Based on existing literature, while higher resolution samples present a high computational expense, especially in the memory of the device used for training, some other studies suggest a search for the appropriate image resolution and model performance [2, 3]. In these studies, researchers compare the trade-offs of certain networks trained at lower versus higher resolutions and evaluate their key performance metrics.

In this study, we propose the implementation and comparison of low-resolution training samples from video-assisted endotracheal intubation to assess the performance of our CNNs and hybrid model, in addition to analyzing the performance of our model(s) when trained from a range of symmetrical resolutions based on previous papers. By doing this, we are analyzing not only common machine learning metrics, but also the trade-off in time that it takes to classify each video as a pass or fail for each resolution. In the context of the machine learning model that we are building, a pass is any video in which an endotracheal intubation is successfully performed, and a fail is any video where the patient is not correctly intubated. In order to classify these videos, the model takes into account factors such as tube location, tube depth, etc., and learns

how to identify when an endotracheal intubation attempt is successful or unsuccessful. Nevertheless, the goal of the experiment being discussed in this report is to use some of the sample resolutions from studies such as “The Effect of Image Resolution on Deep Learning in Radiography” and assess model performance for real-life applications to find an adequate resolution where the model can be fast but also accurate.

Methods

In this experiment, videos with different resolutions (64x64, 128x128, 224x224, 320x320, 600x600) based on the paper by Carl F. et al., “The Effect of Image Resolution on Deep Learning in Radiography,” will be used to train and test our CNN and hybrid model [3]. The results of each model's run at the different resolutions will then be compared using statistical metrics such as AUC, model accuracy, F1-score, and training time. Our primary objective with this test is to compare two models that are trained and tested at the same input dimensions in order to identify the lowest effective resolution.

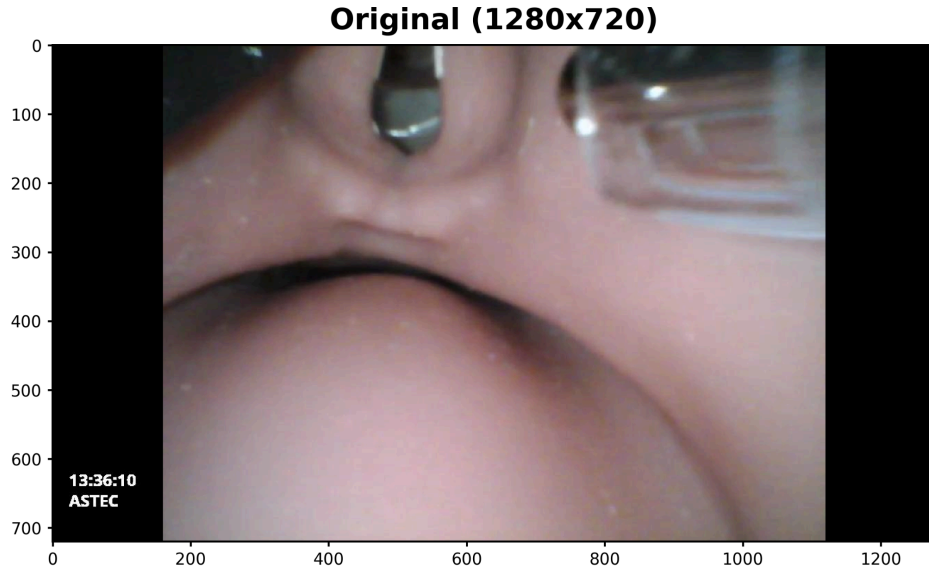


Figure 1: Original Image Resolution

Our dataset currently consists of 50 labeled videos of video-assisted endotracheal intubations; all videos were labeled using polygon labels in the software LabelMe. The small size of the dataset does serve as a limitation in this study and makes it incredibly important to keep the experimental design simplified in order to avoid issues such as overfitting. The 50 videos contain a balanced amount of pass and fail classes that the models classify depending on the spatial or spatio-temporal features learned. The original resolution of each of the videos is 1280x720, shown in Figure 1, in a .avi format, which was taken from the laryngoscopy monitor via USB. To turn the videos into frames, we used the library video-cli, which allowed us to break each video into its respective frames when labeling and when training and testing the models. Overall, this

experiment will test how well the model performs when trained and evaluated at different matched, fixed input sizes, using a 4-fold cross-validation to account for the small sample size.

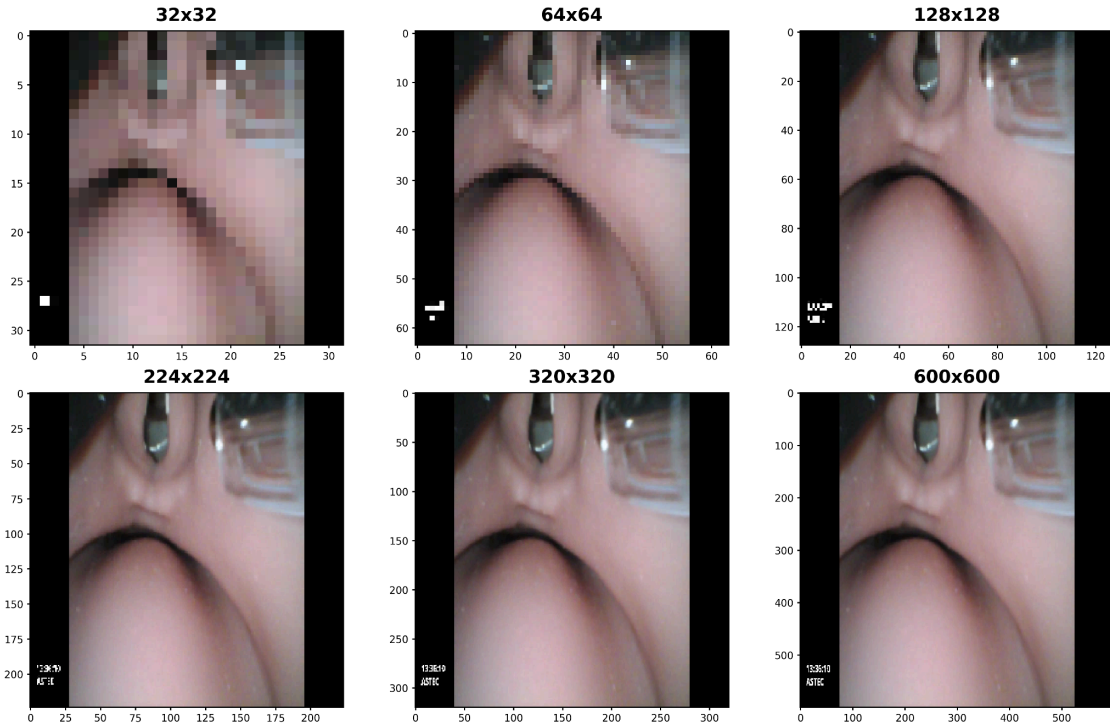


Figure 2: Aspect Ratio Comparison from the Original Image (Figure 1)

As per the implementation and approach of these experiments, we currently plan to: First train our 3DCNN and 2DCNN + SVM models with 50 balanced videos, using 4-fold cross-validation and varying symmetrical resolutions of 64x64, 128x128, 224x224, 320x320, and 600x600 in accordance with Carl F. et al., and as shown in Figure 2 above. Secondly, we will conduct a statistical analysis comparing the mean accuracy, F1-score, AUC, and training time from all folds against each model.

Results

Image Size	Metric	3D-CNN	2D-CNN + SVM	Improvement
64x64	Accuracy	50.00%	83.81%	+33.81%
	Duration	18m 36s	18.19s	~61x faster
	AUC	0.648	0.926	+42.90%
	F1-Score	0.666	0.843	+26.58%
128x128	Accuracy	61.11%	83.38%	+22.27%
	Duration	35m 26s	21.31s	~100x faster
	AUC	0.833	0.926	+11.16%
	F1-Score	0.711	0.843	+18.57%
224x224	Accuracy	61.54%	83.81%	+22.27%
	Duration	1h 28m 02s	23.15s	~228x faster
	AUC	0.747	0.926	+23.96%
	F1-Score	0.739	0.843	+14.07%
320x320	Accuracy	53.85%	83.38%	+29.53%
	Duration	2h 53m 46s	21.32s	~489x faster
	AUC	0.675	0.926	+37.19%
	F1-Score	0.685	0.843	+23.07%
600x600	Accuracy	53.85%	83.38%	+29.53%
	Duration	10h 47m 28s	21.28s	~1,825x faster
	AUC	0.623	0.926	+48.64%
	F1-Score	0.684	0.843	+23.25%

Table 2: Improvement comparison between 3D-CNN & hybrid model.

Based on our results displayed in Table 1, we can come to the general conclusion of the superior performance by our hybrid model utilizing a 2D CNN for feature extraction and SVM for classification. This setup has not only proved to have an overall lower duration time with this model doing 61 times faster for 64x64, 228 times faster for 224x224, and 1,825 times faster for 600x600. The overall accuracy between these two models was compared at an average of 27.48%. Looking at Table 2, we can see these differences more clearly when also evaluating the accuracy of the two models as well. Highlighting that the hybrid model had an average accuracy of 83.55% across all resolutions, with the best resolution determined by duration, accuracy, F1-score, and AUC being 64x64 and 224x224. Meanwhile, as per the 3D-CNN, we saw an exponential increase in training duration as the aspect ratio increased. This difference can be seen in Figure 1. The left side represents the duration of the hybrid model, and the right side

represents the 3D-CNN. Do keep in mind that the overall duration of the 3D-CNN was a maximum of almost 11 hours, therefore, making the vertical maxima different from the hybrid model, which had a max time of 23.15 seconds. Figure 1 below represents the comparison graph of each of these resolutions and models, the 2D CNN + SVM graph (left) is represented in seconds, while the 3D CNN graph (right) is represented in hours.

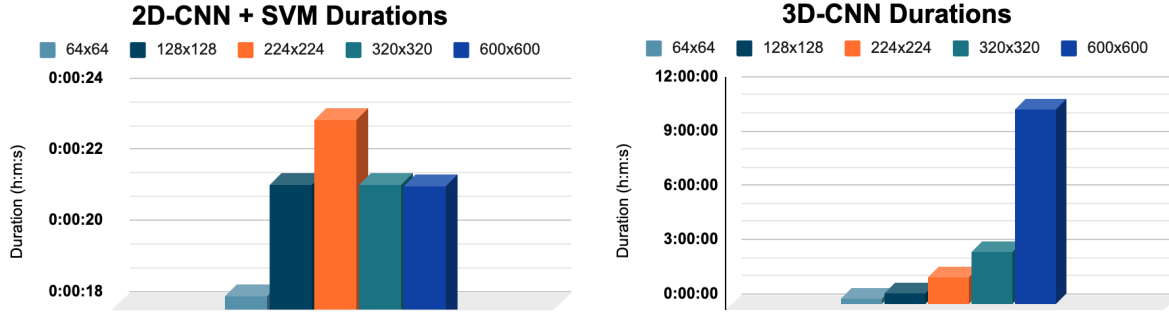


Figure 3: Training Duration comparison between models

As mentioned before, the maximum training time recorded for the 3D-CNN was at 600x600 with a duration of 10 hours, 47 minutes, and 28 seconds. Moreover, we also noticed a bottleneck in accuracy for the 3D-CNN as it tanked at an accuracy of 53.85% after an image size of 320x320. Lastly, we can also compare and analyze that the hybrid model's F1-score and AUC have also improved greatly in comparison to the 3D-CNNs, especially in resolutions such as 64x64 and 600x600. For the sake of coherence, we will only display comparative results of the resolutions that stood out the most, such as 64x64, 600x600, and 224x224 in model accuracy, F1-score, and AUC. However, additional charts will be included in the appendix, showing metric comparison of the other aspect ratios. In summary, when taking into account the training duration of the models and their other statistical metrics, we can determine that the hybrid model excels in all aspect ratios, maintaining a consistent value in all metrics, unlike the 3D-CNN model.

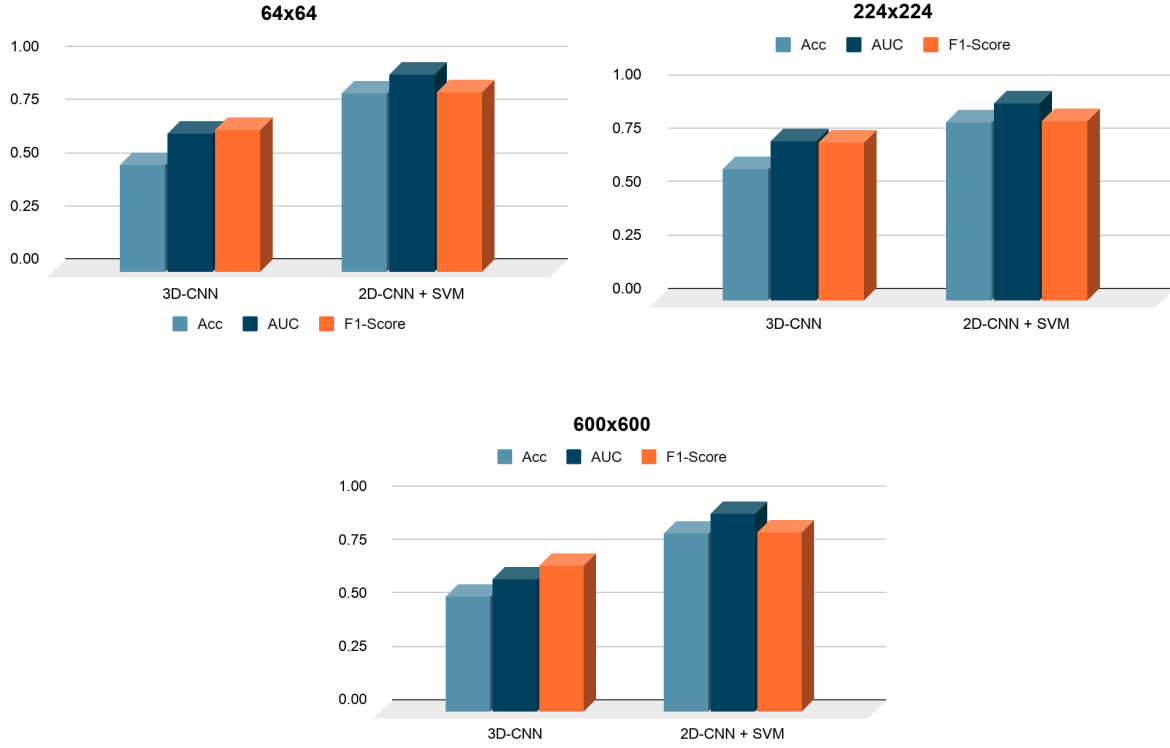


Figure 4: Metric Comparison between Models

Discussion

In relation to the study conducted by Carl F. et al., we see that as the input resolution increases, so does the overall performance, continued by a plateau or even a drop in performance past a resolution of 224x224. While this paper used in reference for this study did not take into account training time for each of these resolutions, we do believe that it is a key metric to take into account. Based on the literature, the field of machine learning continues to face technical challenges as models often need to be retrained with newer or different data. This challenge has brought the integration of different metrics to understand and measure the consumption of training these models, such as power consumption, memory use, and overall training time, to better scale these applications [4]. In relation to our application, being able to understand how long it takes to train these models comes in handy with the usability in a real medical setting, where the patient's outcome is highly relevant in time.

It is important to understand that this experiment does contain certain limitations, such as the small sample size to train these models and the lack of variety in the dataset. The small sample size of 50 videos contains 25 pass and 25 fail attempts, all performed by the same clinician on the same mannequin. This setup does leave this experiment open to overfitting due to the lack of variety in the dataset, as ideally the procedure should have been recorded on different manikins to ensure a wider application. Moreover, while not directly discussed in this paper but supported

by the literature, the SVM classifier is expected to drop in accuracy for this sort of application when more features per data point are added, which in our other experiments had dropped from 92.1% when trained with 20 videos to 83.81% when trained with 50 videos [5]. Meaning that while fast and simpler to deploy and test, this hybrid model might not be the most adequate model for this application in the long term. In relation to this, the overall performance of the 3D CNN was underwhelming, as this was the model that was expected to perform the best based on previous research. Based on our understanding, the 3D CNN could be retrained with different learning rates, batch sizes, epochs, and optimizers, but due to time constraints in the overall project, this won't be the case.

Throughout this experiment we were able to solidify that utilizing the 2D CNN model as a feature extractor, in combination with an SVM classifier, is the best option in comparison to a single 2D CNN and 3D CNN for classifying video-assisted endotracheal intubations. While expected to drop in accuracy as more data is added, the current pipeline does prove that this model is great for prototyping in this application in comparison to the 3D CNN. These results helped determine the direction and importance of the image resolution being fed to this particular model.

Conclusion

In this study we were able to demonstrate that image resolution is a critical factor in balancing performance and computational efficiency for video-based endotracheal intubation classification. This helped demonstrate that while higher resolutions increase computational time and cost, it does not equate to improved performance. In our results, we show that in the case of our 2D CNN + SVM hybrid model, input resolutions such as 64x64 and 224x224 provided the most efficient trade-off between accuracy and efficiency. Moreover, the efficiency of this model showed as it outperformed the 3D-CNN model in multiple metrics, achieving higher accuracy and significantly lower training times. These results support the overall use of hybrid architectures such as this one for practical and efficient solutions in medical diagnosis, more so for ETI classification. However, it is relevant to point out that due to the limited size and variability of the dataset, we do create a limitation and constraint to generalize these results. Furthermore, future work should focus on the expansion of the dataset and fine-tuning of the 3D CNN model for a better comparison.

References

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Appendix

64x64				
Model	Duration (hh:mm:ss)	Accuracy	AUC	F1-Score
3D-CNN	0:18:36	50%	0.648	0.666
2D-CNN + SVM	0:00:18	83.81%	0.926	0.843
128x128				
Model	Duration (hh:mm:ss)	Acc	AUC	F1-Score
3D-CNN	0:35:26	61.11%	0.833	0.711
2D-CNN + SVM	0:00:21	83.38%	0.926	0.843
224x224				
Model	Duration (hh:mm:ss)	Acc	AUC	F1-Score
3D-CNN	1:28:02	61.54%	0.747	0.739
2D-CNN + SVM	0:00:23	83.81%	0.926	0.843
320x320				
Model	Duration (hh:mm:ss)	Acc	AUC	F1-Score
3D-CNN	2:53:46	53.85%	0.675	0.685
2D-CNN + SVM	0:00:21	83.38%	0.926	0.843
600x600				
Model	Duration (hh:mm:ss)	Acc	AUC	F1-Score
3D-CNN	10:47:28	53.85%	0.623	0.684
2D-CNN + SVM	0:00:21	83.38%	0.926	0.843

Table 1: General Metrics Comparison Table between Models

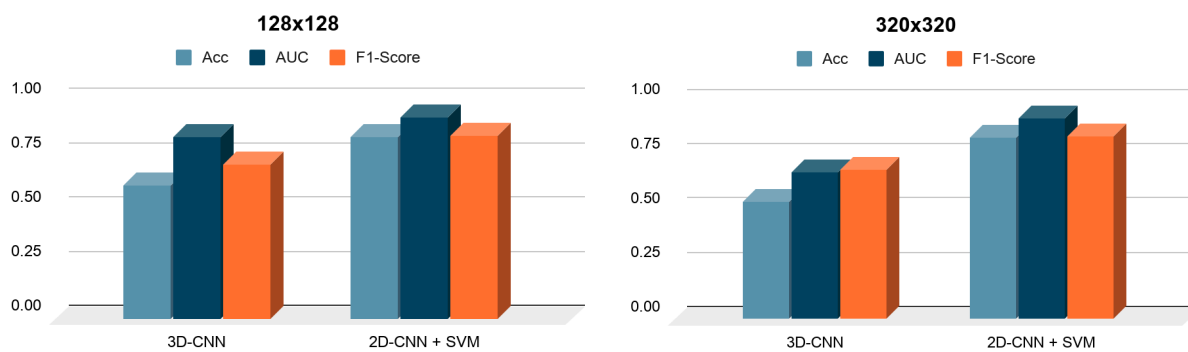


Figure 5: Metric Comparison between Models